

Prove each of the statements in 10-18 by mathematical induction.

5.2.17.
$$\prod_{i=0}^n \left(\frac{1}{2i+1} * \frac{1}{2i+2} \right) = \frac{1}{(2n+2)!} \text{ for every integer } n \geq 0.$$

Proof:

$$P(n) : \prod_{i=0}^n \left(\frac{1}{2i+1} * \frac{1}{2i+2} \right) = \frac{1}{(2n+2)!}$$

Basic step: $P(0)$

LHS:

$$\begin{aligned} \prod_{i=0}^0 \left(\frac{1}{2i+1} * \frac{1}{2i+2} \right) &= \\ \left(\frac{1}{2(0)+1} \right) * \left(\frac{1}{2(0)+2} \right) &= \frac{1}{1} * \frac{1}{2} = \frac{1}{2} \end{aligned}$$

RHS:

$$\begin{aligned} \frac{1}{(2n+2)!} &= \frac{1}{(2(0)+2)!} = \frac{1}{2!} \\ \frac{1}{2!} &= \frac{1}{2 * 1} = \frac{1}{2} \end{aligned}$$

Therefore, $P(0)$ is true.

Inductive step: $P(k+1)$

$$P(k) : \prod_{i=0}^k \left(\frac{1}{2i+1} * \frac{1}{2i+2} \right) = \frac{1}{(2k+2)!}$$

Suppose $P(k)$ is true. We must show inductively that if $P(k)$ is true, then $P(k+1)$ is true.

LHS:

$$\begin{aligned} \prod_{i=0}^k \left(\frac{1}{2i+1} * \frac{1}{2i+2} \right) &= \\ \left(\frac{1}{1} * \frac{1}{2} \right) * \left(\frac{1}{3} * \frac{1}{4} \right) * \dots * \left(\frac{1}{2k+1} * \frac{1}{2k+2} \right) &= \\ \frac{1}{1 * 2 * 3 * 4 * \dots * (2k+1) * (2k+2)} &= \frac{1}{(2k+2)!} \end{aligned}$$

RHS:

$$\frac{1}{(2n+2)!} = \frac{1}{(2k+2)!}$$

Therefore $P(k)$ is true.

Now we can show that $P(k+1)$ is true.

$$P(k+1) : \prod_{i=0}^{k+1} \left(\frac{1}{2i+1} * \frac{1}{2i+2} \right) = \frac{1}{(2(k+1)+2)!}$$

LHS:

$$\begin{aligned} & \prod_{i=0}^{k+1} \left(\frac{1}{2i+1} * \frac{1}{2i+2} \right) \\ & \left(\frac{1}{1} * \frac{1}{2} \right) * \left(\frac{1}{3} * \frac{1}{4} \right) * \dots * \left(\frac{1}{2k+1} * \frac{1}{2k+2} \right) * \left(\frac{1}{2(k+1)+1} * \frac{1}{2(k+1)+2} \right) = \\ & = \prod_{i=0}^k \left(\frac{1}{2i+1} * \frac{1}{2i+2} \right) * \left(\frac{1}{2(k+1)+1} * \frac{1}{2(k+1)+2} \right) = \\ & \frac{1}{1 * 2 * 3 * 4 * \dots * (2(k+1)+1) * (2(k+1)+2)} = \frac{1}{(2(k+1)+2)!} \end{aligned}$$

RHS:

$$\frac{1}{(2n+2)!} = \frac{1}{(2(k+1)+2)!}$$

Therefore, $P(k+1)$ is true.

□

Solution: $\prod_{i=0}^n \left(\frac{1}{2i+1} * \frac{1}{2i+2} \right) = \frac{1}{(2n+2)!}$ for every integer $n \geq 0$.